

Network Security Project

By

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As a part of studying Cyber Security in John Bryce academy I was introduced with the concept of Active Directory and Windows Servers on which they ran. The concept was new and intriguing with many new concepts of securing work environment in different organizations. We started as how it looks in the inside creating a new active directory, new users and new rule policies. The 2nd part was about the view of the attacker, how to enumerate the domain, find the domain controller, laminate movement between different computers to find stored data that could affect the organization or the user who have left unsecured stuff. All this orchestrating by our teacher Doron Zohar had lead us to create an automatization to check the domain, map him and find his vulnerabilities and present them to the client for further investigation and fixing the breaches that was found on them.

This project was partially hard in perspective of finding the correct syntax from the specific tool you were using and specific grep's so the output will server us in the next upcoming level of enumerating or trying to exploit onwards.

Thanks to Doron Zohar and Ai chatbots Deepseek and Claude for teaching, explaining and showing how to handle this uneasy project.

Few Pictures From What you about to experience:

```

kali@kali: ~/Desktop/Bash_scripting/NS_Project
Session Actions Edit View Help
50ff1241b3b1fcff9f90b3e62273d7d3ed1a2c640602230724569df08deaa6d4217dd72923023516d5eee99ffbd4db3772af767df7fbb9599c21dec4
d569821b2acf52e7905852a52b71061b18abfee88c4cf8c2d7e682d2e746815d100
✓ Tickets Extracted, Moving To John!

🔑 Checking Pass lists
✓ User Pass list detected, Proceeding with /home/kali/Desktop/Bash_scripting/NS_Project/passcheck.txt
Using default input encoding: UTF-8
Loaded 5 password hashes with 5 different salts (krb5asrep, Kerberos 5 AS-REP etype 17/18/23 [MD4 HMAC-MD5 RC4 / PBKDF2
Will run 4 OpenMP threads
Press Ctrl-C to abort, or send SIGUSR1 to john process for status
Warning: Only 9 candidates left, minimum 32 needed for performance.
TmaAebCaoPao4 ($krb5asrep$23$Barbara@NETSEC.LOCAL)
ReinaAgustine55 ($krb5asrep$23$Matthew@NETSEC.LOCAL)
2g 0:00:00:00 DONE (2025-12-06 07:50) 66.66g/s 300.0p/s 1500c/s 1500C/s root..ReinaAgustine55
Use the "--show" option to display all of the cracked passwords reliably
Session completed.
$krb5asrep$23$Barbara@NETSEC.LOCAL:TmaAebCaoPao4
$krb5asrep$23$Matthew@NETSEC.LOCAL:ReinaAgustine55

2 password hashes cracked, 3 left
✓ Attack Completed, Report Saved in test1/netsec.local_cracked.txt

✓ Finished Exploit Phase!

📄 Summarising Everything into PDF File!
[ 2 pages * 1 copy ] left in test1/report
8 lines were wrapped
✓ Report Pdf Created in test1 Named report.pdf!

⌚ Script Complete in : 373 seconds

```

```

(kali@kali)~[~/Desktop/Bash_scripting/NS_Project]
$

```

```

kali@kali: ~/Desktop/Bash_scripting/NS_Project
Session Actions Edit View Help
| IDs: CVE:CVE-2017-0143
| Risk factor: HIGH
| A critical remote code execution vulnerability exists in Microsoft SMBv1
| servers (ms17-010).
|
| Disclosure date: 2017-03-14
| References:
| https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2017-0143
| https://blogs.technet.microsoft.com/msrc/2017/05/12/customer-guidance-for-wannacrypt-attacks/
| https://technet.microsoft.com/en-us/library/security/ms17-010.aspx
|_ smb-vuln-ms10-061: NT_STATUS_ACCESS_DENIED

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 152.09 seconds
✓ Basic Exploit Complete! Saving Report to test1/netsec.local_vuln.txt

🔥 INTERMEDIATE EXPLOIT 🔥
✓ test1/netsec.local_users.txt Present! Beginning Work!
🔑 Checking For Passlist to spray with
✓ User Pass list detected, Proceeding with /home/kali/Desktop/Bash_scripting/NS_Project/passcheck.txt
SMB 192.168.112.160 445 WIN-A8PBA9GL4GE [+] netsec.local\hack3r:1234567aA
SMB 192.168.112.160 445 WIN-A8PBA9GL4GE [+] netsec.local\King:Dpwc8h2yx! (Pwn3d!)
SMB 192.168.112.160 445 WIN-A8PBA9GL4GE [+] netsec.local\pentestme:Dpwc8h2yx!
SMB 192.168.112.160 445 WIN-A8PBA9GL4GE [+] netsec.local\Barbara:TmaAebCaoPao4
SMB 192.168.112.160 445 WIN-A8PBA9GL4GE [+] netsec.local\Mary:C4uE1wN5q
SMB 192.168.112.160 445 WIN-A8PBA9GL4GE [+] netsec.local\Mark:C4uE1wN5q
SMB 192.168.112.160 445 WIN-A8PBA9GL4GE [+] netsec.local\Matthew:ReinaAgustine55
SMB 192.168.112.160 445 WIN-A8PBA9GL4GE [+] netsec.local\Richard:C4uE1wN5q
✓ Attack Completed, Report Saved in test1/netsec.local_spray.txt

🔥 ADVANCED EXPLOIT 🔥
✓ User List Detected!
🔑 Extracting Kerberos tickets ...

```

```
kali@kali: ~/Desktop/Bash_scripting/NS_Project
Session Actions Edit View Help
636/tcp open  tcpwrapped
3268/tcp open  ldap          Microsoft Windows Active Directory LDAP (Domain: netsec.local, Site: Default-First-Site-Name)
3269/tcp open  tcpwrapped
5985/tcp open  http          Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_http-stored-xss: Couldn't find any stored XSS vulnerabilities.
|_http-dombased-xss: Couldn't find any DOM based XSS.
|_http-server-header: Microsoft-HTTPAPI/2.0
|_http-csrf: Couldn't find any CSRF vulnerabilities.
MAC Address: 00:0C:29:F2:DA:90 (VMware)
Service Info: Host: WIN-A8PBA9GL4GE; OS: Windows; CPE: cpe:/o:microsoft:windows
```

Host script results:

```
|_smb-vuln-ms10-054: false
|_smb-vuln-ms17-010:
|  VULNERABLE:
|    Remote Code Execution vulnerability in Microsoft SMBv1 servers (ms17-010)
|    State: VULNERABLE
|    IDs:  CVE:CVE-2017-0143
|    Risk factor: HIGH
|    A critical remote code execution vulnerability exists in Microsoft SMBv1
|    servers (ms17-010).
|
|    Disclosure date: 2017-03-14
|    References:
|      https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2017-0143
|      https://blogs.technet.microsoft.com/msrc/2017/05/12/customer-guidance-for-wannacrypt-attacks/
|      https://technet.microsoft.com/en-us/library/security/ms17-010.aspx
|_smb-vuln-ms10-061: NT_STATUS_ACCESS_DENIED
```

Service detection performed. Please report any incorrect results at <https://nmap.org/submit/> .

Nmap done: 1 IP address (1 host up) scanned in 152.09 seconds

√ Basic Exploit Complete! Saving Report to test1/netsec.local_vuln.txt

```
kali@kali: ~/Desktop/Bash_scripting/NS_Project
Session Actions Edit View Help
krttgt
DefaultAccount
Guest
Administrator
√ Finished Enumirating Users! Saved into test1/netsec.local_users.txt

♥ Starting Enumerating Password Policy!
[+] Dumping password info for domain:
Minimum password length: 7
Password history length: 24
Maximum password age: 41 days 23

Password Complexity Flags: 000001
Domain Refuse Password Change: 0
Domain Password Store Cleartext: 0
Domain Password Lockout Admins: 0
Domain Password No Clear Change: 0
Domain Password No Anon Change: 0
Domain Password Complex: 1

Minimum password age: 1 day 4
Reset Account Lockout Counter: 30 minutes
Locked Account Duration: 30 minutes
Account Lockout Threshold: None
Forced Log off Time: Not Set
√ Finished Enumirating Password Policies! Saved into test1/netsec.local_posspol.txt

👤 Starting Enumerating Domain Admins Group!
[+] Enumerated
netsec.local\King
netsec.local\Administrator
√ Finished Enumirating Domain Admins Group! Saved into test1/netsec.local_domadgrp.txt
```

```
kali@kali: ~/Desktop/Bash_scripting/NS_Project
Session Actions Edit View Help
\\192.168.112.160\NETLOGON:
warning: Couldn't get details for share: NT_STATUS_ACCESS_DENIED
Anonymous access: <none>

√ Share Enumeration Complete! Proceeding to smb-os enum!

smb-os-discovery:
OS: Windows Server 2016 Standard Evaluation 14393 (Windows Server 2016 Standard Evaluation 6.3)
Computer name: WIN-A8PBA9GL4GE
NetBIOS computer name: WIN-A8PBA9GL4GE\x00
Domain name: netsec.local
Forest name: netsec.local
FQDN: WIN-A8PBA9GL4GE.netsec.local
System time: 2025-12-06T04:47:43-08:00

√ SMB-Os Enumeration Complete! Proceeding to smb protocols enum!

smb-protocols:
dialects:
NT LM 0.12 (SMBv1) [dangerous, but default]
2:0:2
2:1:0
3:0:0
3:0:2
3:1:1

√ SMB Protocols Enumeration Complete!
√ Finished nmap scripts, Saved into test1/nmap_script.txt

√ User Credentials Found Continuing
√ Password Found Continuing
===== ADVANCED ENUMERATION =====
📌 Starting Enumerating Shares!

kali@kali: ~/Desktop/Bash_scripting/NS_Project
Session Actions Edit View Help
√ Version Scan Complete! Saved into test1/192.168.112.2.txt
√ Version Scan Complete! Saved into test1/192.168.112.160.txt
√ Version Scan Complete! Saved into test1/192.168.112.254.txt
√ Version Scan Complete! Saved into test1/192.168.112.134.txt

🔴 The Domain Ip is : 192.168.112.160
🟢 The DHCP Ip is : 192.168.112.254

===== INTERMEDIATE ENUMERATION =====
🟢 The Next Ip's Have 21 on:
192.168.112.1
192.168.112.254

🟢 The Next Ip's Have 22 on:

🟢 The Next Ip's Have 389 on:
192.168.112.160

🟢 The Next Ip's Have 445 on:
192.168.112.160

🟢 The Next Ip's Have 3389 on:

🟢 The Next Ip's Have 5985 on:
192.168.112.160

√ Scanning for Ports completed,Saved to test1/open_ports.txt, Proceeding

📌 Starting Nmap Script For Shared Folders, SMB Os and Protocols
```



```
kali@kali: ~/Desktop/Bash_scripting/NS_Project
Session Actions Edit View Help
=====Entering Enumeration Phase=====
+ Please Select The Level of Enumeration you want
x Remember you can use -h for more info on the enumerations options provided
* A.Basic Enumeration
* B.Intermediate Enumeration
* C.Advanced Enumeration. note works only with providet domain user and password
C

+ You Have Chosen Advanced Enumerating, Beginning Work!

===== BASIC ENUMERATION =====
v Version Scan Complete! Saved into test1/192.168.112.1.txt
v Version Scan Complete! Saved into test1/192.168.112.2.txt
v Version Scan Complete! Saved into test1/192.168.112.160.txt
v Version Scan Complete! Saved into test1/192.168.112.254.txt
v Version Scan Complete! Saved into test1/192.168.112.134.txt
v The Domain Ip is : 192.168.112.160
```

```
kali@kali: ~/Desktop/Bash_scripting/NS_Project
Session Actions Edit View Help
=====Entering Enumeration Phase=====
+ Please Select The Level of Enumeration you want
x Remember you can use -h for more info on the enumerations options provided
* A.Basic Enumeration
* B.Intermediate Enumeration
* C.Advanced Enumeration. note works only with providet domain user and password
C

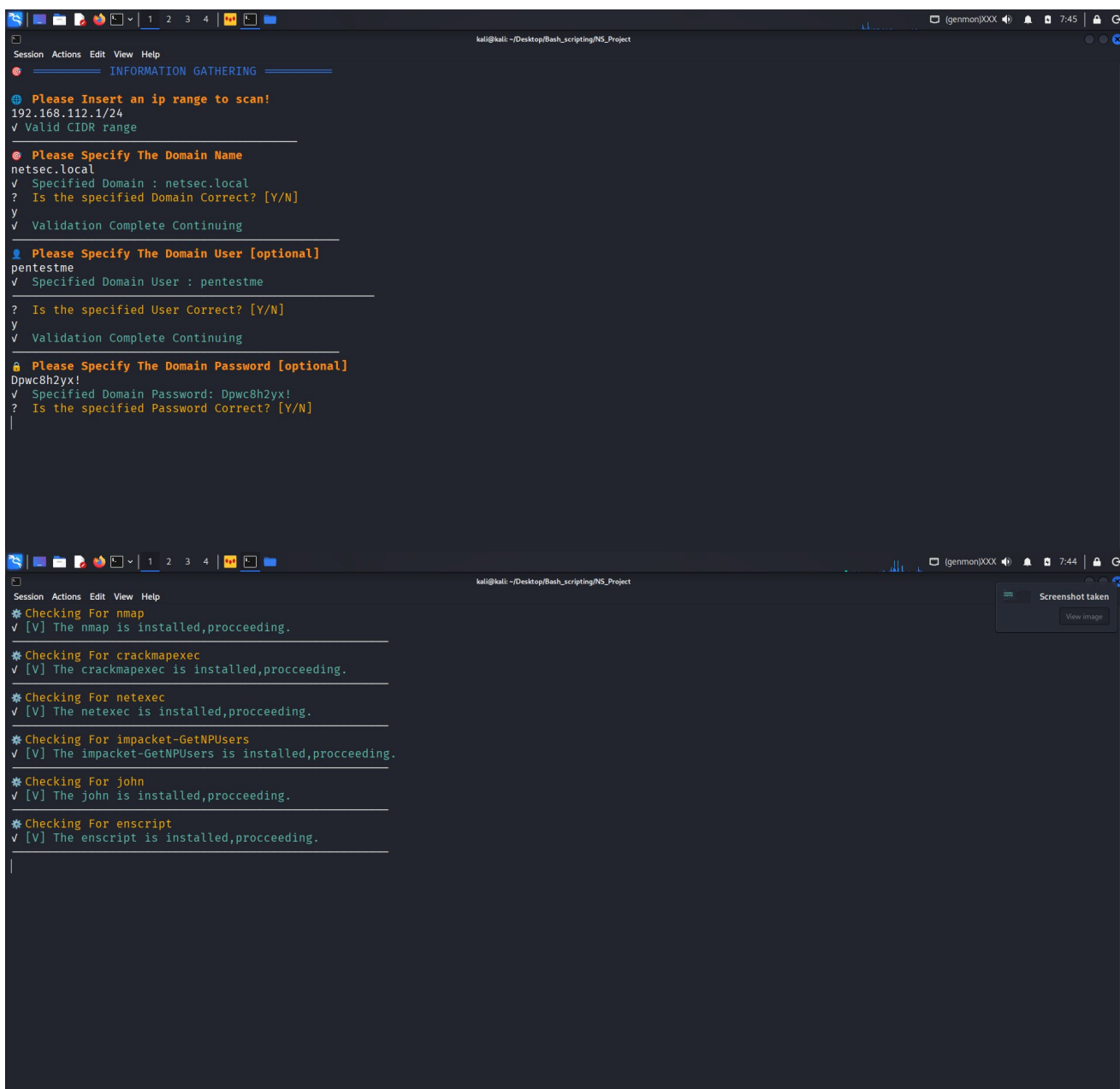
+ You Have Chosen Advanced Enumerating, Beginning Work!

===== BASIC ENUMERATION =====
v Version Scan Complete! Saved into test1/192.168.112.1.txt
```

```
kali@kali: ~/Desktop/Bash_scripting/NS_Project
Session Actions Edit View Help
=====Entering Scanning Phase=====
+ Please Select The Level of Scanning you want
x Remember you can use -h for more info on the scanning options provided
* A.Basic Scanning
* B.Intermediate Scanning
* C.Advanced Scanning
a
+ You Have Chosen Basic Scanning, Beginning Work!
v Scanning Complete! Saving to test1/nmap_output.txt

===== SAVING CREDENTIALS =====
v Saving Providet Credentials as Providet in Created Folder

Tested Ip Range : 192.168.112.1/24
Tested Domain Name : netsec.local
Tested User Name : pentestme
Tested Password : Dpwc8h2yx!
```





* Starting Domain Mapper ...